

GCCM Results: Urban Heat Drivers in Ouagadougou

What is GCCM?

Geographical Convergent Cross Mapping (Gao et al. 2023, *Nature Communications*) tests whether two spatial variables share underlying dynamics. It goes beyond correlation by asking: does the spatial pattern of one variable contain enough information to reconstruct the other?

Each pixel is described by the mean values of concentric rings around it. If the ring-pattern of variable Y can predict variable X, that is evidence that X influences Y's spatial structure.

GCCM Spatial Embedding: What E and tau control

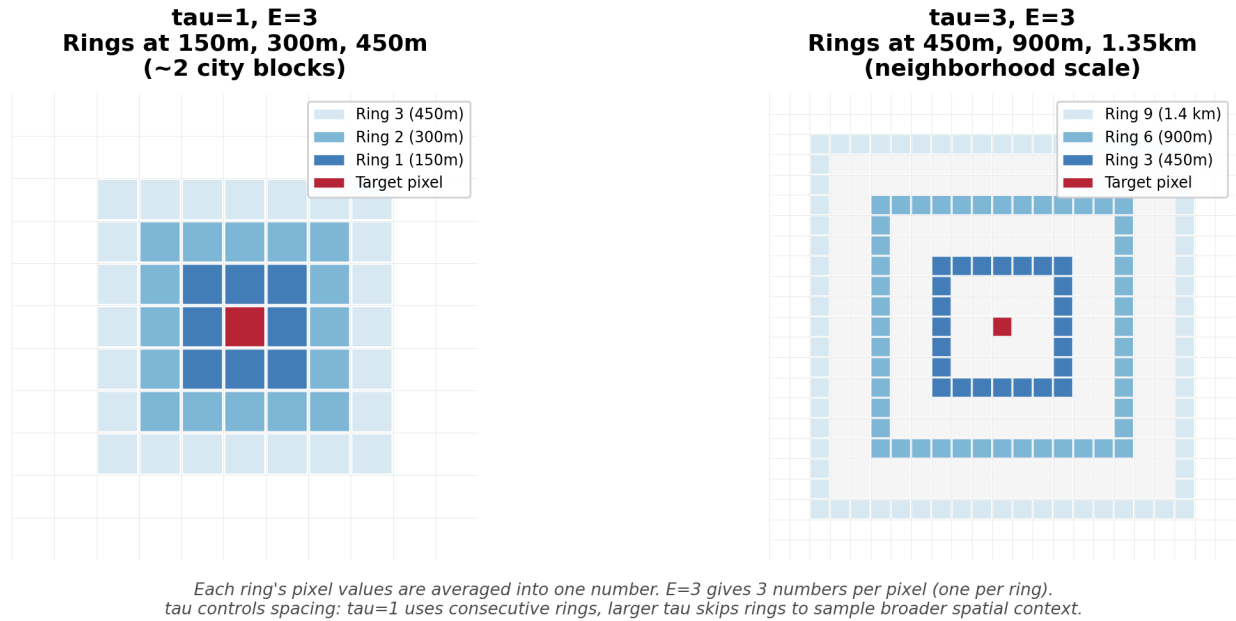


Figure 1: GCCM ring diagram

We tested 8 surface properties against Land Surface Temperature (LST) in Ouagadougou at 150m resolution (~25,000 valid pixels).

Predictor	Source	Type
built_density	ESA WorldCover 2021	90m kernel density
green_density	ESA WorldCover 2021	90m kernel density
distance_to_water	JRC / OSM	Static
distance_to_roads	OSM	Static
DEM	Copernicus GLO-30	Static (control variable)
NDBI	Sentinel-2, 2024 Mar-May	Spectral index
BSI	Sentinel-2, 2024 Mar-May	Spectral index
NDVI	Sentinel-2, 2024 Mar-May	Spectral index

Target: **LST** from Landsat 8/9, 2022-2024 Mar-May median (different sensor and time window from Sentinel-2 predictors).

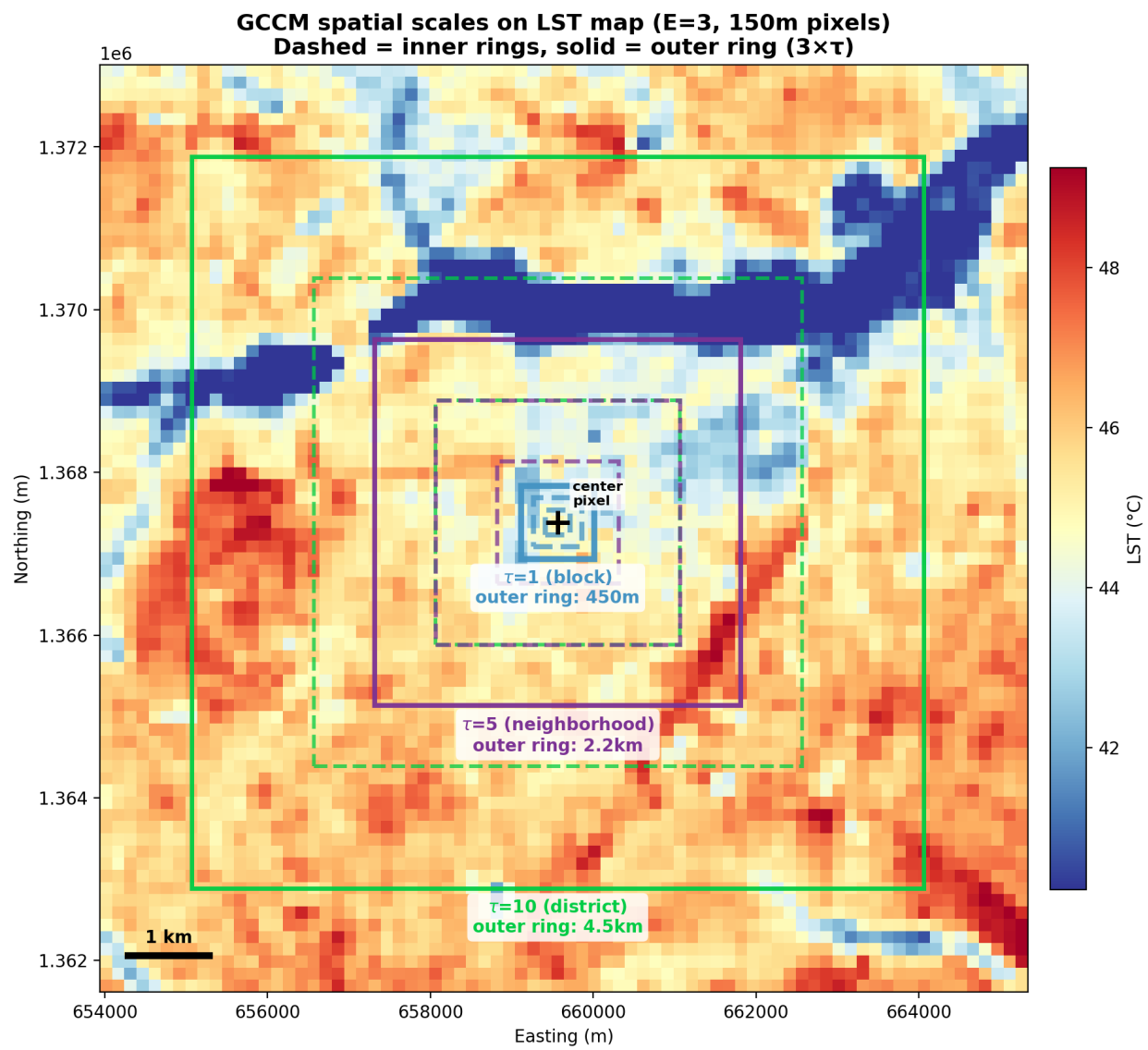


Figure 2: Scale comparison

All variables are coupled with LST

The signature of coupling in GCCM is **convergence**: prediction skill (ρ) increases as the method uses more spatial context (larger library size). Every variable shows clear convergence in both directions across all parameter configurations we tested.

E=3, tau=1 (block scale, 450m reach):

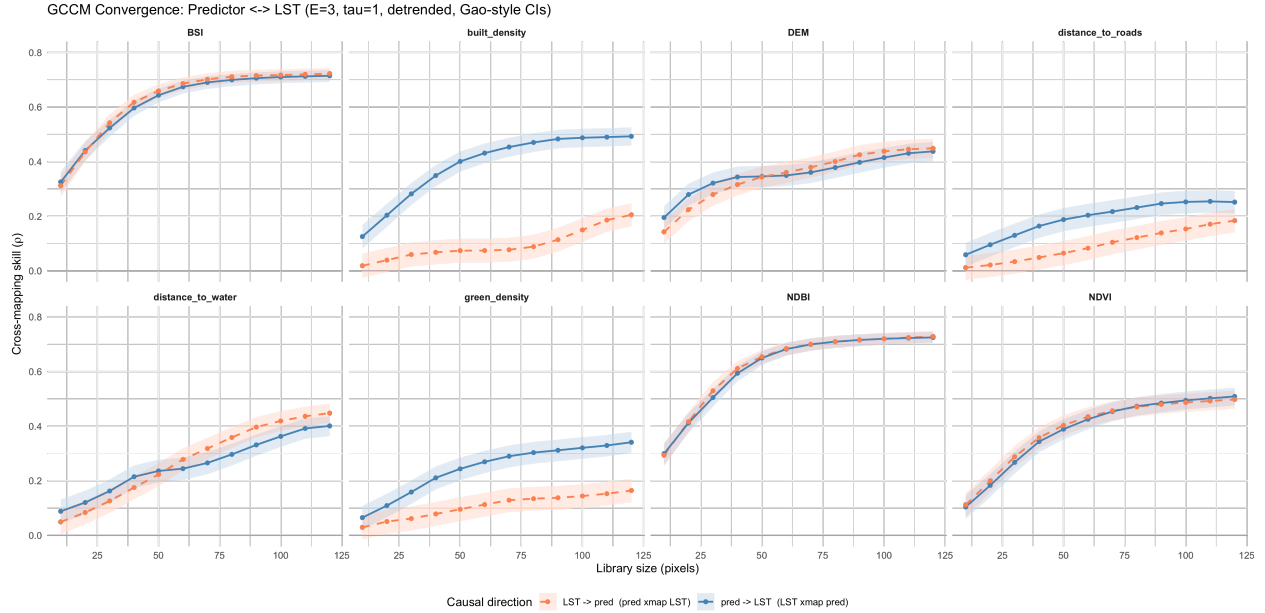


Figure 3: Convergence E3 tau1

E=3, tau=5 (neighborhood scale, 2.25km reach):

E=3, tau=10 (district scale, 4.5km reach):

Reminder: E = number of rings, τ = spacing between rings. With $E=3$ at 150m resolution, $\tau=1$ uses rings at 150m/300m/450m (block scale), $\tau=5$ uses rings at 750m/1.5km/2.25km (neighborhood scale), while $\tau=10$ uses rings at 1.5km/3km/4.5km (district scale).

All three configurations show lines going up from left to right for every variable. This convergence is the primary evidence of dynamical coupling between each surface property and LST.

Interpretation of convergence:

1. **Lines going up** = convergence. This is the key test. It means that as the method sees more spatial context, it gets better at predicting. That increase is the evidence of dynamical coupling. A flat or decreasing line would mean no coupling.
2. **Separation between the two lines** = directional evidence. If one line is consistently higher, that direction has stronger cross-mapping, which in theory indicates causal direction. In practice, most of our variables show the lines close together or overlapping (see note about confidence intervals below).
3. **Final ρ value** (right end of each line) = the coupling strength at the largest library size. This is the number reported in the summary tables.

About the confidence intervals: The shaded ribbons are 95% CIs computed via Fisher-z transform, following Gao et al. (2023). $N = 2000$ prediction points.

Where the CI ribbons for the two directions **do not overlap**, there is statistically significant evidence that one direction is stronger. Where they overlap, we cannot distinguish the directions.

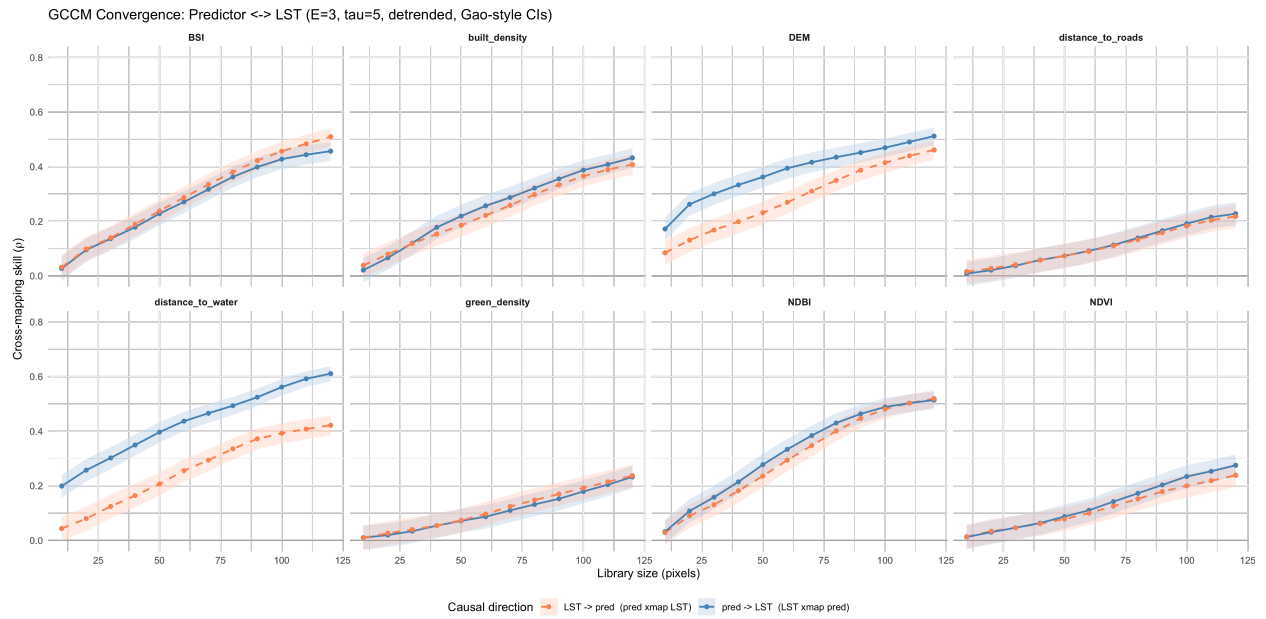


Figure 4: Convergence E3 tau5

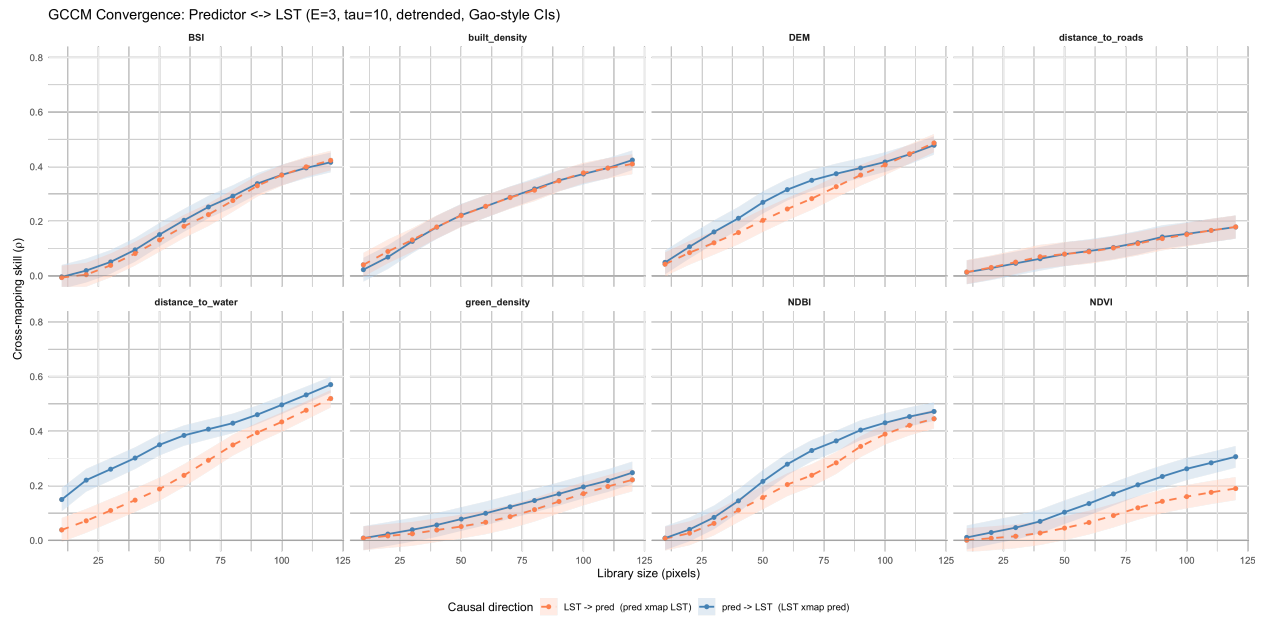


Figure 5: Convergence E3 tau10

How strong is the coupling?

Our rho values are comparable to or stronger than published GCCM results:

Published benchmarks (Gao et al. 2023, global/regional scales):

Variable pair	rho (X causes Y)	rho (Y causes X)
Elevation causes Population	~0.64	~0.36
Precipitation causes NPP	0.37-0.83	varies
Temperature causes NPP	0.17-0.50	varies

Published benchmarks (Yeboah et al. 2025, GCCM on LULCC in Chongqing):

Variable pair	rho (X causes UHI)
Built-up causes UHI	0.32-0.63
Bare land causes UHI	0.25-0.42
Vegetation, waterbodies cause UHI	negative (cooling)

Gao’s benchmarks are at coarse resolution (county/global scale). Yeboah’s are at 30m Landsat resolution, closer to our 150m setup, making them a more direct comparison. Our values are comparable or stronger across the board.

Our results (E=3, tau=1):

Predictor	rho (pred causes LST)	rho (LST causes pred)
NDBI	0.73	0.73
BSI	0.71	0.72
NDVI	0.51	0.50
built_density	0.49	0.21
DEM	0.44	0.45
distance_to_water	0.40	0.45
green_density	0.34	0.16
distance_to_roads	0.25	0.18

Our results (E=3, tau=5):

Predictor	rho (pred causes LST)	rho (LST causes pred)
distance_to_water	0.61	0.42
NDBI	0.51	0.52
DEM	0.51	0.46
BSI	0.46	0.51
built_density	0.43	0.41
NDVI	0.28	0.24
green_density	0.23	0.24
distance_to_roads	0.23	0.22

Our results (E=3, tau=10):

Predictor	rho (pred causes LST)	rho (LST causes pred)
distance_to_water	0.57	0.52
DEM	0.48	0.49
NDBI	0.47	0.45
built_density	0.42	0.41
BSI	0.42	0.42
NDVI	0.31	0.19
green_density	0.25	0.22
distance_to_roads	0.18	0.18

All values are statistically significant ($p < 0.05$). The 95% CIs, computed via Fisher-z transform with $N = 2000$ prediction points following Gao et al. (2023), exclude zero for all variables in both directions.

Can we determine causal direction?

GCCM tests direction by comparing forward rho (pred causes LST, measured by LST cross-mapping pred) vs reverse rho (LST causes pred, measured by pred cross-mapping LST). If one direction is significantly stronger, that suggests causal direction.

The bar chart below shows forward and reverse rho for each variable, with the asymmetry (difference) annotated. Green = pred causes LST (expected direction), red = LST causes pred.

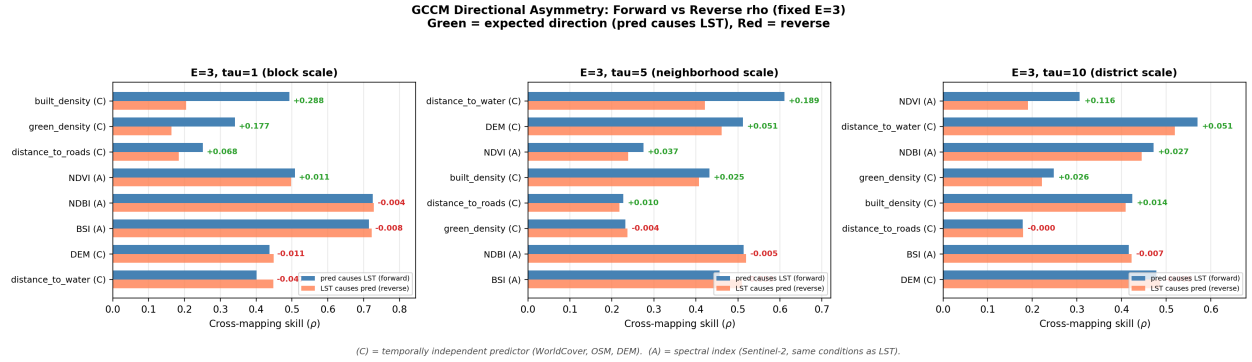


Figure 6: Asymmetry bars

Labels: (C) = temporally independent predictors (WorldCover, OSM, DEM; acquired independently from LST). (A) = spectral indices (Sentinel-2, 2024 Mar-May; different sensor and partially different time window from Landsat LST 2022-2024 Mar-May, but both capture similar seasonal conditions). All three panels use fixed E=3; only tau differs.

Most asymmetries are small, and the ranking changes across panels. built_density goes from the strongest positive asymmetry (+0.29) at tau=1 to nearly zero (+0.01) at tau=10. distance_to_water shows the strongest signal at tau=5 (+0.19), where it clearly separates from the pack. DEM (our control variable, known direction: DEM causes LST) shows the wrong sign at tau=1 and tau=5, but is near zero at tau=10.

Results are sensitive to parameter choices (E and tau)

We ran four configurations for all 8 variables to test how the choice of embedding dimension (E) and spatial lag (tau) affects the results. Comparing adjacent configs isolates one parameter at a time:

- $E=3/\tau=1$ vs $E=3/\tau=5$ vs $E=3/\tau=10$: τ effect (E held constant)
- $E=3/\tau=10$ vs simplex- $E/\tau=10$: E effect (τ held constant)

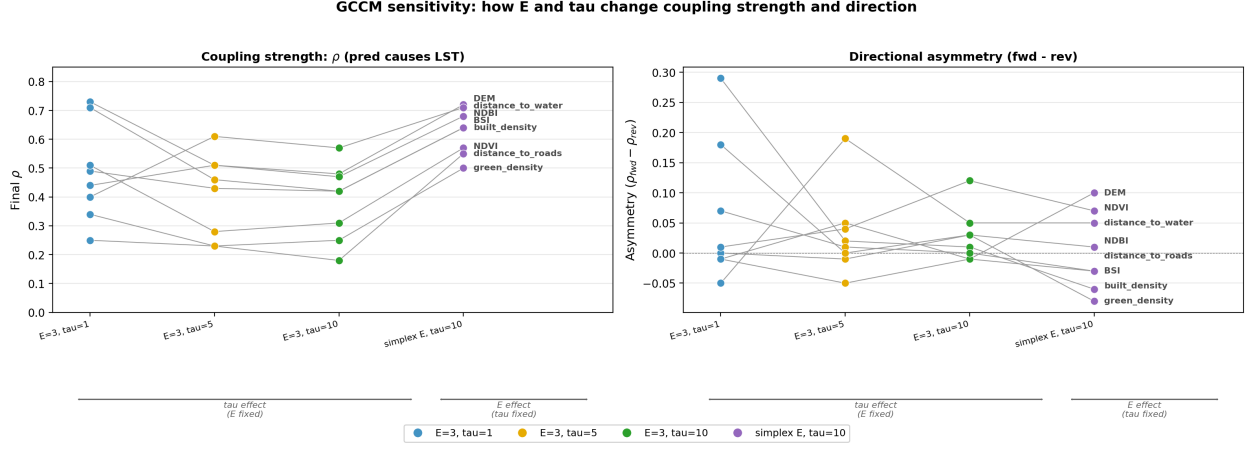


Figure 7: Sensitivity dot plot

Same data, colored by predictor to trace individual variables across configs:

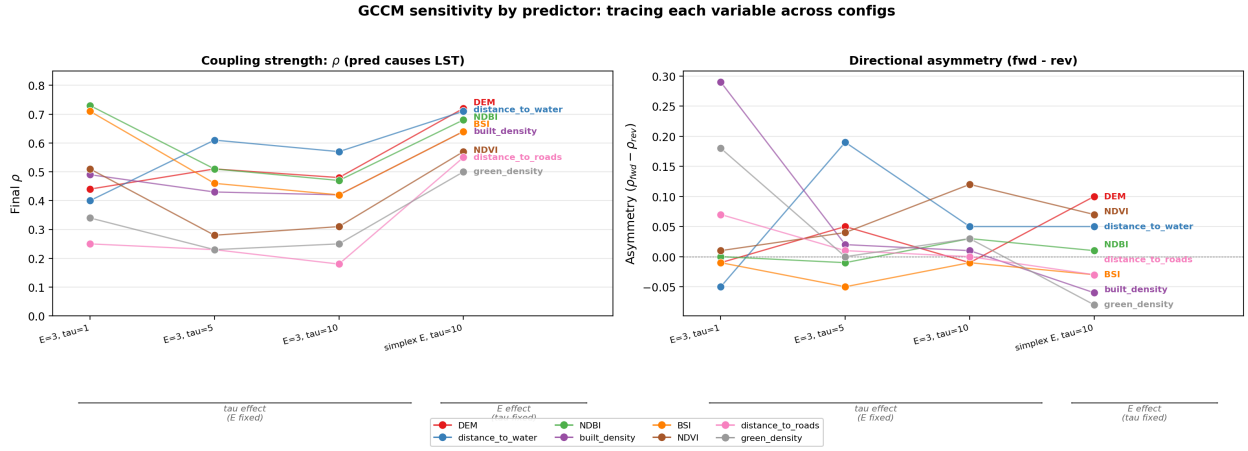


Figure 8: Sensitivity by feature

Left panel (coupling strength): Rho values shift across configs, but all variables remain significantly coupled with LST in every configuration. Coupling is robust to parameter choice.

Right panel (asymmetry): The directional signal is not stable. For example, built_density has asymmetry +0.29 at $E=3/\tau=1$ (strongly “pred causes LST”), but flips to -0.06 at simplex- $E/\tau=10$. green_density similarly flips from +0.18 to -0.08. DEM, our control variable (known direction: DEM causes LST), only shows the correct positive asymmetry in the simplex- $E/\tau=10$ config.

Predictor	E=3, $\tau=1$		E=3, $\tau=5$		E=3, $\tau=10$		simplex E, $\tau=10$	
	fwd	asym	fwd	asym	fwd	asym	fwd	asym
DEM	0.44	-0.01	0.51	+0.05	0.48	-0.01	0.72 (E=9)	+0.10

Predictor	E=3, tau=1		E=3, tau=5		E=3, tau=10		simplex E, tau=10
distance_to_water	0.40	-0.05	0.61	+0.19	0.57	+0.05	0.71 (E=9)
NDBI	0.73	-0.00	0.51	-0.01	0.47	+0.03	0.68 (E=10)
built_density	0.49	+0.29	0.43	+0.02	0.42	+0.01	0.64 (E=12)
BSI	0.71	-0.01	0.46	-0.05	0.42	-0.01	0.64 (E=11)
NDVI	0.51	+0.01	0.28	+0.04	0.31	+0.12	0.57 (E=10)
distance_to_roads	0.25	+0.07	0.23	+0.01	0.18	-0.00	0.55 (E=12)
green_density	0.34	+0.18	0.23	-0.00	0.25	+0.03	0.50 (E=10)

fwd = rho (pred causes LST). *asym* = fwd minus rev (positive = expected direction).

Key takeaway: Coupling (are these variables linked to LST?) is robust across all parameter choices. Direction (which drives which?) is not.

Why direction is hard at this resolution

The plot below shows Pearson correlation between each variable and a spatially shifted copy of itself, as a function of distance (in pixels, where 1 pixel = 150m). For each variable: how similar is a pixel's value to pixels 1, 2, 3, ... steps away? A value of 1.0 means identical, 0.0 means no relationship. This is a property of the data, independent of GCCM.

The x-axis is lag distance in pixels, which maps directly onto GCCM's tau parameter: when tau=1, GCCM uses rings at lags 1, 2, 3 (read off the left side of the plot where autocorrelation is highest). When tau=10, the rings are at lags 10, 20, 30 (further right, where autocorrelation has dropped).

At lag=1 (150m), all variables have $r > 0.87$. This means neighboring pixels carry nearly the same information. With tau=1, the three rings at 150m, 300m, and 450m are all highly redundant. Both the forward and reverse cross-mapping directions can achieve high rho simply because nearby spatial structure is so predictable, not because of a causal signal.

This is a known limitation of CCM-family methods: when coupling is strong, both directions become significant, and the small difference between them is sensitive to parameter choices (Ye et al. 2015).

No published GCCM study has been conducted at sub-km resolution, so there is no established benchmark for how directional asymmetry should behave at this spatial scale.

Simplex selection of E

The standard EDM approach is to let simplex projection choose E automatically: try E=2, 3, ... 15 and pick whichever gives the best nearest-neighbor prediction skill. In temporal data from chaotic systems, prediction skill peaks at the true system dimension and then declines and that peak is meaningful.

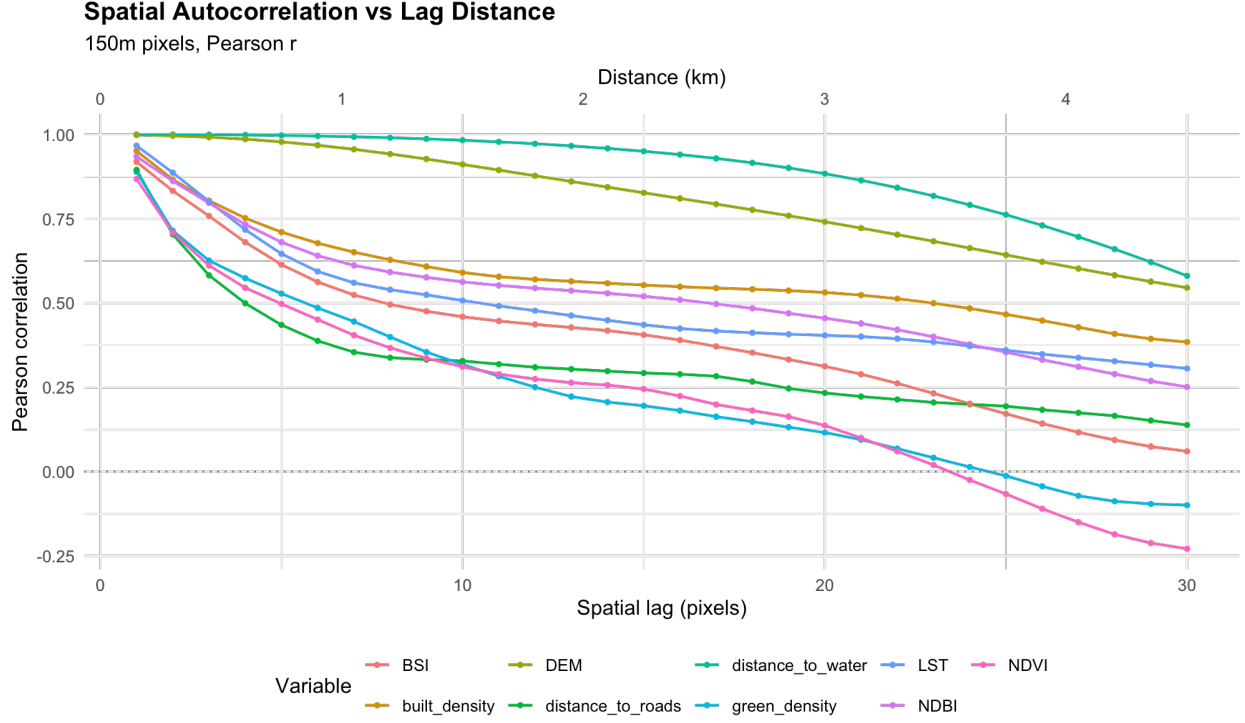


Figure 9: Spatial autocorrelation

In spatially autocorrelated raster data, this might not be meaningful. Each additional ring adds a tiny bit of extra information (redundant but not perfectly so), so prediction skill keeps climbing with no peak. Simplex just returns whatever the maximum of the search range is.

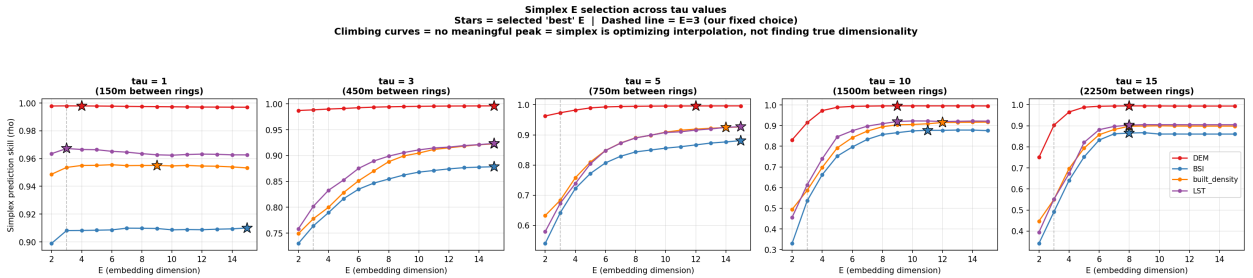


Figure 10: Simplex E curves

Each panel shows one tau value. Lines = simplex prediction skill (rho) at each E. Stars = the “best” E that simplex selected. Dashed gray = E=3 (our fixed choice).

- **tau=1:** DEM and LST flatten early (E=3-4), but BSI still climbs to E=15.
- **tau=3:** All four variables climb monotonically — every one maxes out at E=15.
- **tau=5:** Same pattern — 3 of 4 hit E=14-15.
- **tau=10, 15:** Curves show more diminishing returns, but still no clean peak.

These are likely not meaningful dimensionality estimates, since simplex is chasing marginal interpolation gains in spatial data.

This matters for our GCCM: At high E (9-15) and high tau, each pixel’s state vector captures a huge chunk of the city. Both the LST cloud and predictor cloud end up encoding the same city-wide gradient,

so cross-mapping works equally well in both directions and the asymmetry signal disappears. At $E=3$, the clouds encode more local spatial structure where causal differences can emerge.

We fix $E=3$ following Gao et al. (2023), who used $E=2-3$ in their published examples.

The choice of E and τ for future analyses should probably be more theoretically driven, as opposed to data driven, where chosen E and τ are based on what geographic extent we want to cover and what we assume should be the relevant extent of influence of a feature (e.g. maybe green spaces have very local extent of influence occurring on the scale of 1-100 meters, while DEM has a much wider extent of influence on LST occurring on 100-1000 meters).

For example, $\tau=3$ with $E=8$ gives 8 rings spaced 450m apart, spanning from 450m to 3.6km — covering block-to-neighborhood scale variation with finer spatial resolution than $\tau=10/E=3$, while sampling a comparable total extent.

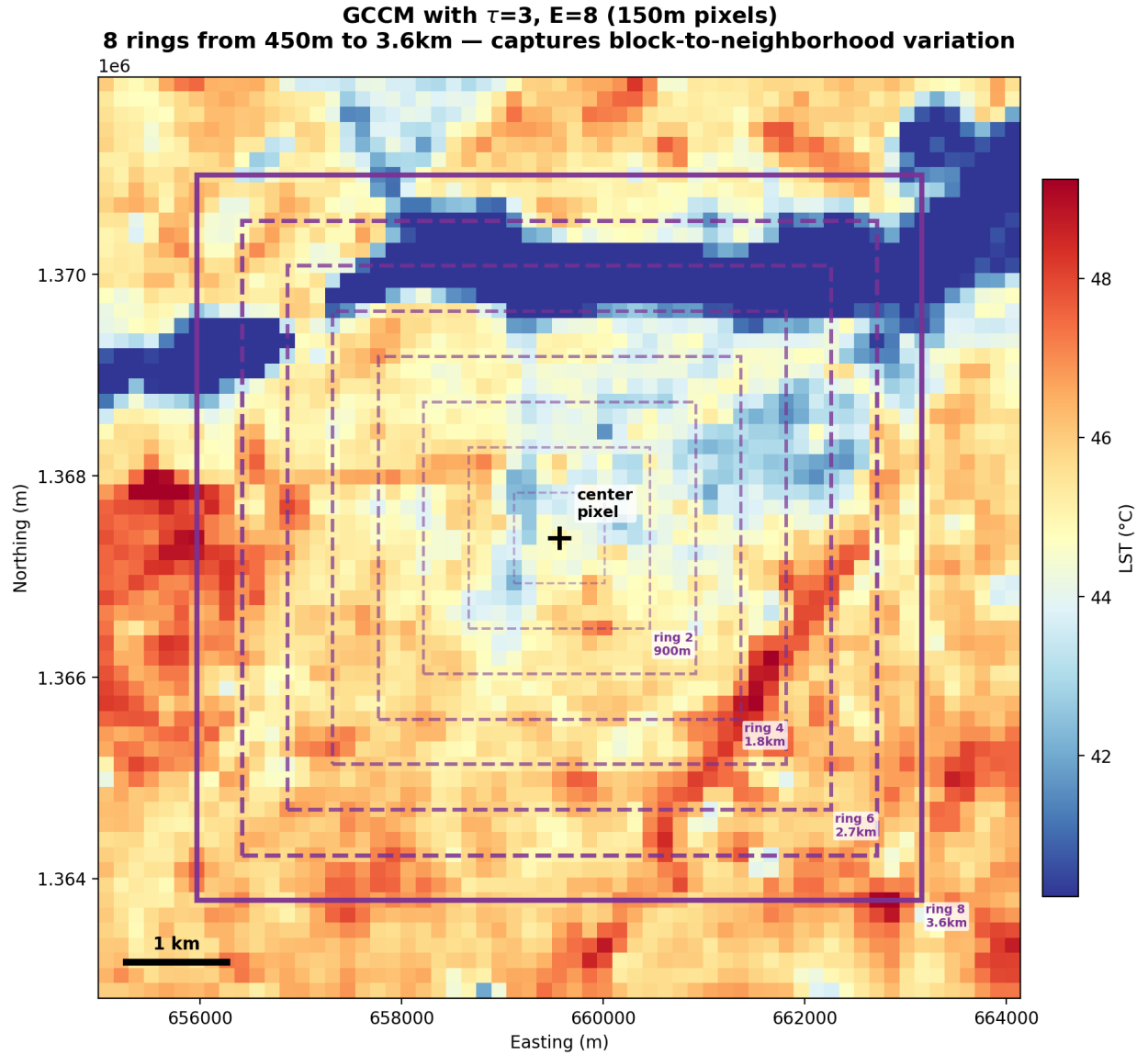


Figure 11: Scale $\tau=3$ $E=8$

A configuration like this could be a reasonable theoretically-motivated starting point: the inner rings capture

local effects (e.g. park cooling within a few blocks) while the outer rings pick up neighborhood-scale structure (e.g. proximity to the reservoir), all without collapsing into a single city-wide gradient the way high-E/high-tau combinations do.

Unfortunately, this configuration wasn't run yet.

What we can conclude

Robust findings (consistent across all parameter configurations):

- All 8 surface properties show significant dynamical coupling with LST
- Coupling strength is comparable to published GCCM benchmarks
- The built environment (built_density, distance_to_roads, distance_to_water) is genuinely linked to the temperature field, beyond simple correlation
- Spectral indices (NDBI, BSI) show the strongest coupling with LST

Open questions:

- Causal direction cannot be reliably determined at 150m resolution
- The method has not been validated at sub-km scales in the literature

Multi-scale analysis with fixed E=3 and varying tau. Fixing E=3 (matching Gao's protocol) removes one source of instability, while varying tau lets us ask at which spatial scale each variable most strongly drives LST. Each tau corresponds to a real urban scale:

Config	Ring distances	Outer reach	Urban meaning	Status
tau=1, E=3	150m, 300m, 450m	450m	~1-2 city blocks	Done
tau=5, E=3	750m, 1.5km, 2.25km	2.25km	Neighborhood	Done
tau=8, E=3	1.2km, 2.4km, 3.6km	3.6km	Large neighborhood / small district	Planned
tau=10, E=3	1.5km, 3km, 4.5km	4.5km	District	Done

At tau=5, distance_to_water shows the strongest asymmetry of any variable in any configuration (+0.19), suggesting water proximity may exert its strongest causal influence at the neighborhood scale. However, the DEM control also shows positive asymmetry (+0.05) at tau=5, meaning directional results at this scale should be interpreted with caution.

References

- Gao, B. et al. (2023). Causal inference from cross-sectional earth system data with geographical convergent cross mapping. *Nature Communications*, 14, 5875.
- Ye, H. et al. (2015). Distinguishing time-delayed causal interactions using convergent cross mapping. *Scientific Reports*, 5, 14750.
- Lyu, W. (2026). spEDM: Spatial Empirical Dynamic Modeling. R package. doi:10.32614/CRAN.package.spEDM
- Yeboah, E. et al. (2025). A causal investigation of land use and land cover change on emerging urban heat island footprints in a mid-latitude region. *Environment, Development and Sustainability*.